



Cyber Security & Digital Forensics

CSE - 4207

Lecture - 01

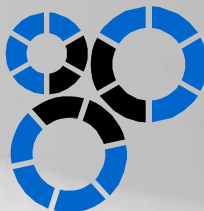
Dr. Md. Alam Hossain

Professor

Jashore University of Science & Technology

Outline

- The Security Mindset
- Meet the Adversary
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- Key Security Concepts
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- Aspects of Security
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- Model for Network Security
- Model for Network Access Security



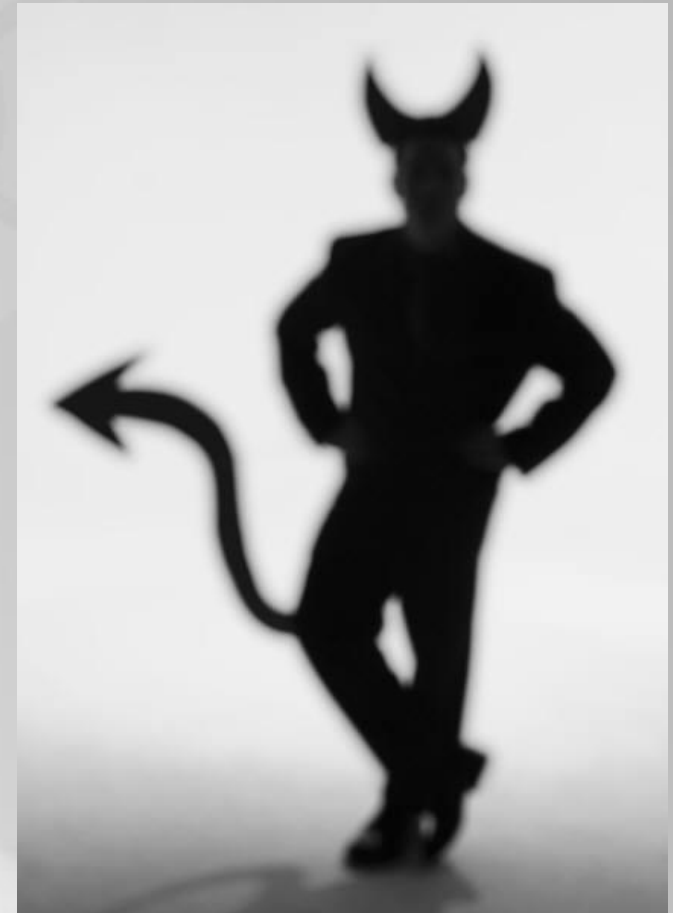
◉ The Security Mindset

- “Security requires a particular mindset. Security professionals --at least the good ones --see the world differently. They can't walk into a store without noticing how they might shoplift. They can't use a computer without wondering about the security vulnerabilities. They can't vote without trying to figure out how to vote twice. They just can't help it.”

— Bruce Schneier

◉ Meet the Adversary

- “Computer security studies how systems behave in the presence of an adversary.”
- The adversary - the attacker/ the bad guy
- An intelligence that actively tries to cause the system to misbehave.



◉ Know Your Enemy

- The art of war teaches us to rely not on the likelihood of the enemy's not coming, but on our own readiness to receive him; not on the chance of his not attacking, but rather on the fact that we have made our position unassailable.

—*The Art of War, Sun Tzu*



Thinking Like an Attacker

- Attackers try to find the weakest part of a system because it is easier to break.
- They look for the smallest mistake or weakness that can help them enter the system.
- Security often depends on some assumptions. Attackers check if those assumptions are actually wrong or false.
- Attackers also think in unusual ways. They are not limited by the system designer's ideas and may try unexpected methods to attack the system.

Find the Code





- What are some security systems you interact with in everyday life?

1. Password Authentication System

- Used to protect accounts such as email, social media, banking, and mobile devices.
- Requires a username and password to verify the user's identity.

2. Biometric Security System

- Uses physical characteristics like fingerprints, face recognition, or iris scans.
- Example: Unlocking smartphones using fingerprint or Face ID.

3. Firewall

- Protects computers and networks by monitoring and blocking unauthorized network access.
- Example: Windows Firewall protecting a computer from suspicious connections.

4. Antivirus Software

- Detects, prevents, and removes malware such as viruses, worms, and trojans.
- Example: Windows Defender scanning files for threats.

5. Encryption Systems

- Protect sensitive information by converting data into an unreadable format.
- Example: HTTPS websites, encrypted messaging apps, and online banking.

6. Two-Factor Authentication (2FA)

- Provides an extra layer of security by requiring a second verification step.
- Example: Entering an OTP after entering a password.

7. Access Control Systems

- Allow only authorized people to enter restricted areas.
- Example: Smart card access in offices, fingerprint scanners in buildings.

8. CCTV Surveillance Systems

- Monitor and record activities for security purposes.
- Example: Security cameras in banks, shops, and homes.

9. Spam and Phishing Protection Systems

- Detect and block suspicious emails or messages.
- Example: Email services automatically moving spam emails to the spam folder.

10. Secure Payment Systems

- Protect financial transactions using encryption and authentication.
- Example: ATM cards, mobile banking, and online payment gateways.

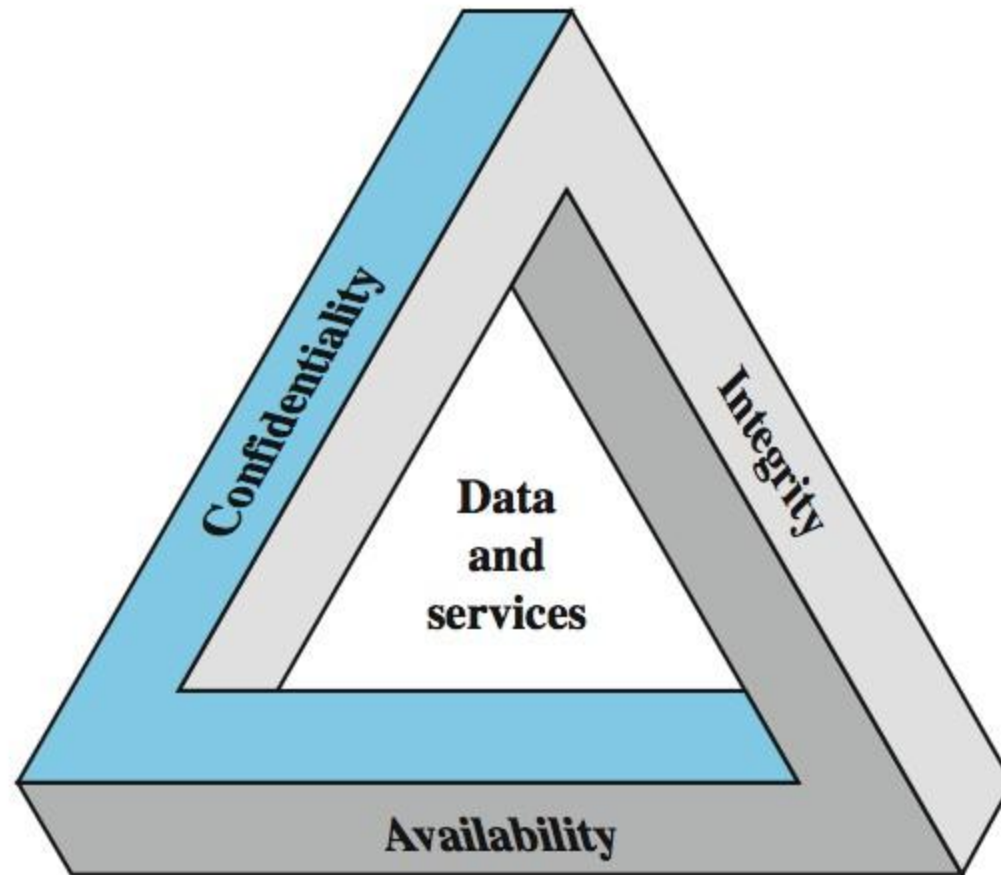


Security and Standards Organizations

- National Institute of Standards & Technology (NIST)
- Internet Society (ISOC)
- International Telecommunication Union
Telecommunication Standardization Sector (ITU-T)
- International Organization for Standardization (ISO)



Key Security Concepts





Levels of Impact From a Security Branch

Low Impact

Causes small problems or limited damage.

System still works normally most of the time.

Example: A user account password is exposed but quickly recovered.

Moderate Impact

Causes noticeable damage or disruption.

Some important data or services may be affected.

Example: A company website goes down for several hours due to an attack.

High Impact

Causes serious damage, major data loss, or long-term disruption.

Can affect many users, money, or important operations.

Example: A hospital system is hacked and patient data is stolen.



Examples of Security Requirements

- Confidentiality – Student grades should only be seen by authorized people like students and teachers.
- Integrity – Patient information must stay correct and should not be changed illegally.
- Availability – Authentication services should always be accessible so users can log in when needed.



Computer Security Challenges

1. **Not simple** – Computer security is complex and difficult to manage.
2. **Must consider potential attacks** – Security systems must be designed while thinking about possible hacker attacks.
3. **Procedures can be counter-intuitive** – Some security rules may seem unusual or inconvenient to users.
4. **Involves algorithms and secret information** – Security uses encryption, passwords, and confidential data.
5. **Must decide where to deploy mechanisms** – It is important to choose the correct place to apply security protections.
6. **Battle of wits between attacker and admin** – Attackers try to break security while administrators try to stop them.
7. **Benefits are not noticed until failure happens** – People often realize the importance of security only after an attack occurs.
8. **Requires regular monitoring** – Security systems need continuous checking and updates.
9. **Too often an afterthought** – Many systems add security only after development is finished.
10. **Regarded as an impediment to using systems** – Some users think security measures make systems harder to use.

◉ Aspects of Security

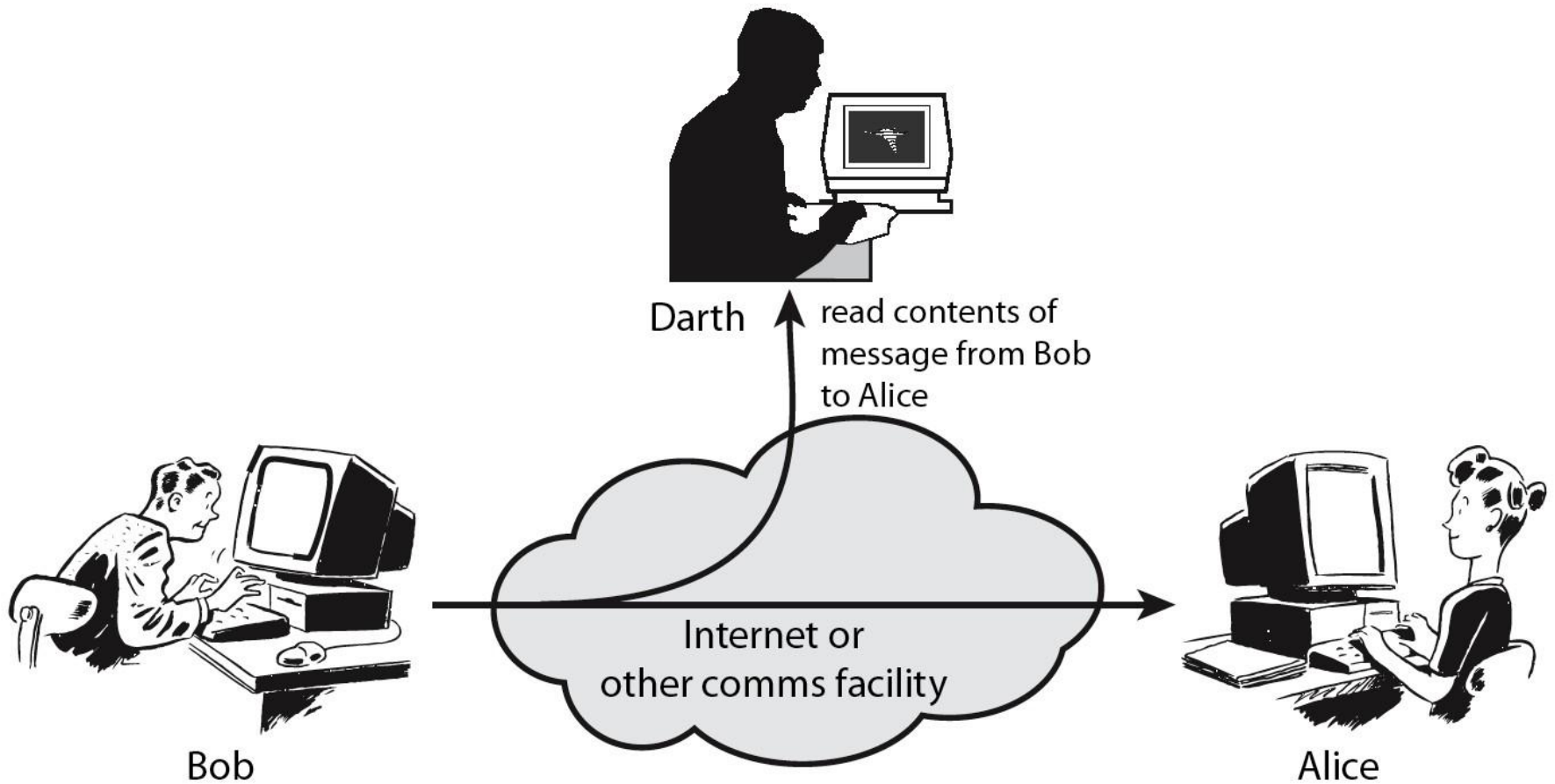
- consider 3 aspects of information security:
 - security attack
 - security mechanism
 - security service
- note terms
 - **threat** – Something that may cause harm to a system or break security.
 - **attack** – A deliberate action to break security or damage a system.

Passive & Active Attack

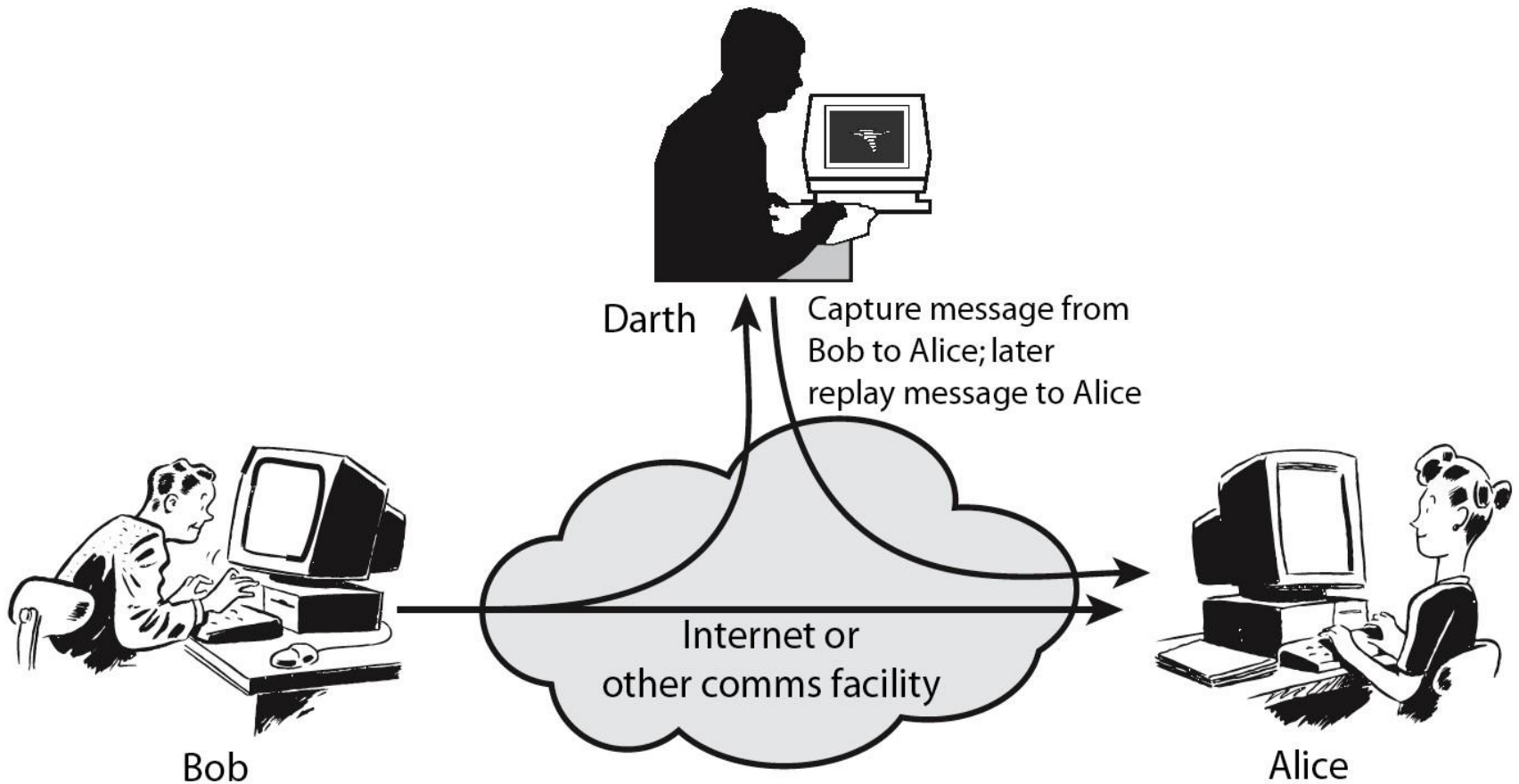
A passive attack happens when an attacker secretly observes or collects information without changing anything in the system. Its main goal is to steal or monitor data, such as listening to network traffic or reading confidential information.

An active attack happens when an attacker changes, damages, or disrupts the system or data. Examples include hacking websites, modifying files, or launching DoS attacks to stop services from working.

○ Passive Attacks



◉ Active Attacks

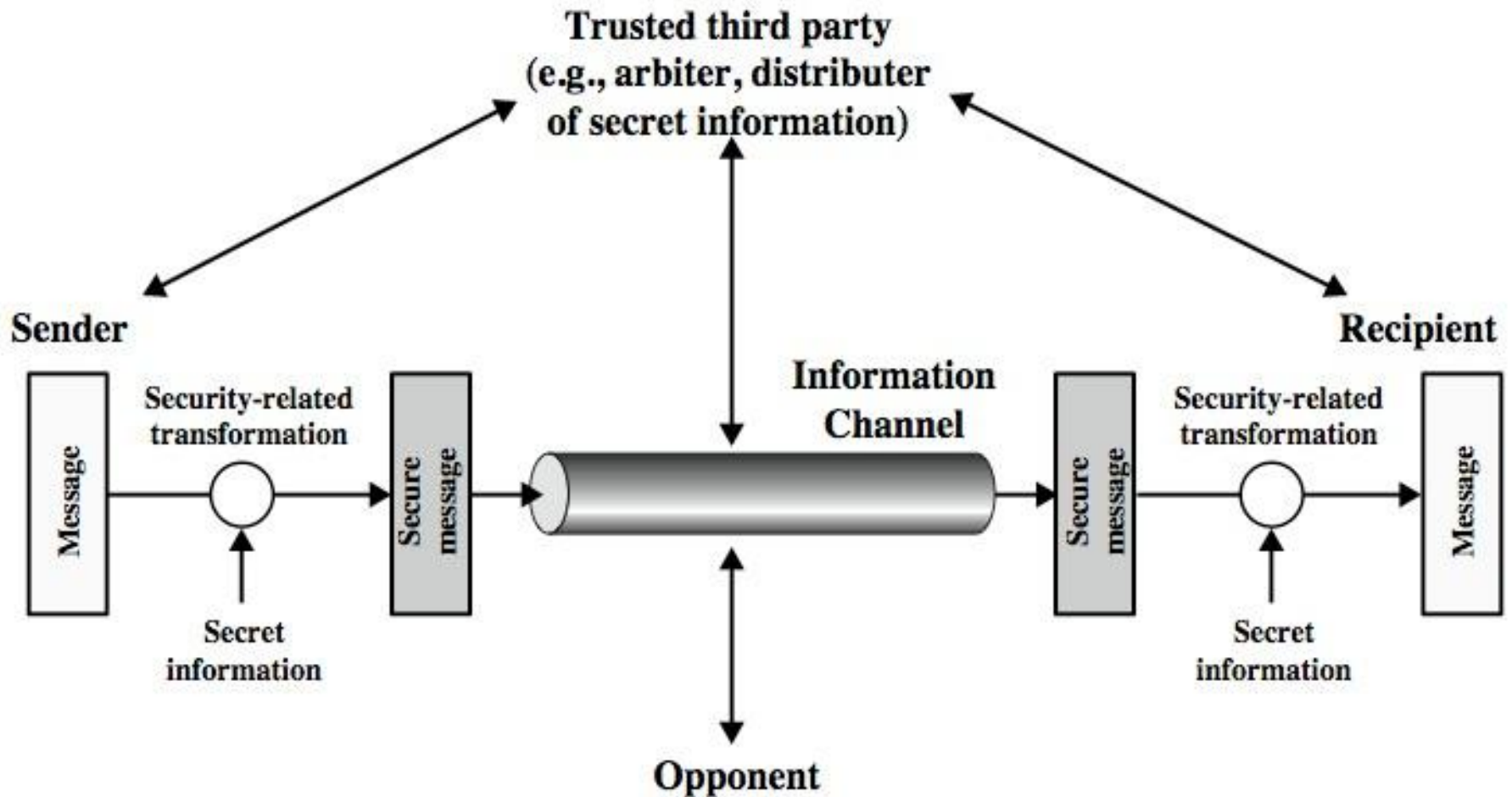


◉ Security Services

- X.800:
 - “According to X.800, a security service is a service that protects computer systems and data during communication.”
- RFC 2828:
 - “According to RFC 2828, a security service is a system service that provides specific protection for system resources and information.”



Model for Network Security

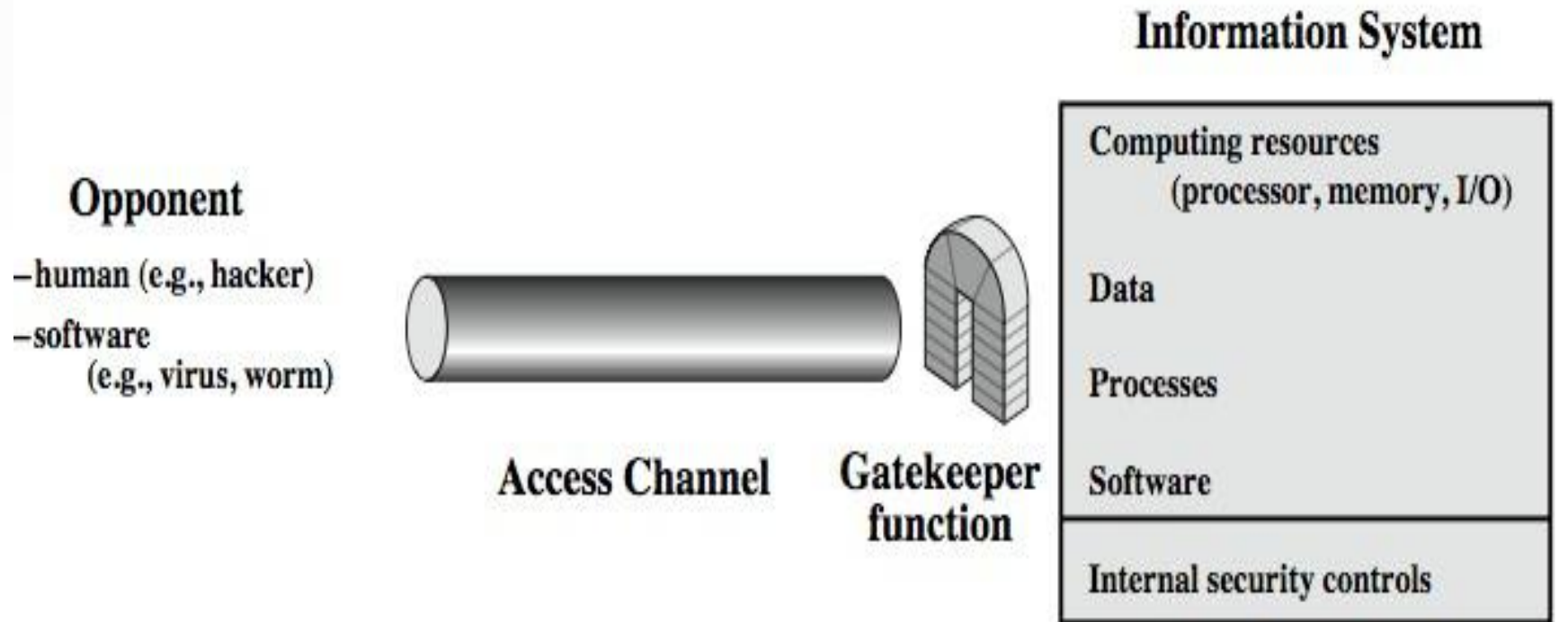




Model for Network Security

- Using this model requires us to:
 1. Design a suitable algorithm for the security transformation
 2. Generate the secret information (keys) used by the algorithm
 3. Develop methods to distribute and share the secret information
 4. Specify a protocol enabling the principals to use the transformation and secret information for a security service

Model for Network Access Security



Model for Network Access Security

- Using this model requires us to:
 1. Select appropriate gatekeeper functions to identify users
 2. Implement security controls to ensure only authorized users access designated information or resources



Thank You

